

SESSION 1: INTRODUCTION TO FX CARRY

Earning Returns from Interest Rate Differentials in the Foreign Exchange Market

1. What is FX Carry?

- Borrow in a low-interest-rate currency.
- Invest in a high-interest-rate currency.
- Earn the interest-rate differential ('carry').
- Returns also depend on exchange-rate movements.

2. Economic Logic

- Covered Interest Parity (CIP) determines forward prices.
- Carry strategies exploit the empirical failure of Uncovered Interest Parity (UIP).

3. Key Formulae

Covered Interest Parity (USD/Foreign quotation):

$$F = S \times (1 + r_{\text{foreign}}) / (1 + r_{\text{USD}})$$

Uncovered Interest Parity (UIP):

$$(1 + r_{\text{USD}}) = (1 + r_{\text{foreign}}) \times E[S(t+1)] / S(t)$$

Carry Signal (Forward Discount):

$$\text{Carry} \approx (F - S) / S \approx r_{\text{foreign}} - r_{\text{USD}}$$

Return Decomposition:

$$\text{Total Return} \approx \text{Carry Return} + \text{Currency Return}$$

$$\text{Carry Return} \approx \text{Interest-Rate Differential}$$

$$\text{Currency Return} \approx (S(t+1) - F) / S(t)$$

4. Example

- Funding currencies: JPY, CHF
- Investment currencies: AUD, NZD, MXN
- Positive return if FX depreciation is smaller than the carry earned.